

RHYTHM BHETWAL

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Education

National Institute of Technology, Meghalaya | CSE | CGPA: 9.49 | 2024 – 2028

Skills

Languages: Python, C++, C, SQL

Data Engineering & Automation: Airflow, Prefect, Docker, AWS (EC2, S3), API Development (FastAPI, Flask), RESTful Integration

Databases: PostgreSQL, SQLite

Tools & Frameworks: Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, Plotly, OpenCV, Selenium, BeautifulSoup, Scikit-learn, TensorFlow, Keras

Domains: Machine Learning, Deep Learning, NLP, Time Series, Computer Vision, Web Scraping, Data Visualization

Projects

- **Logical Inconsistency Based Hallucination Detection System (Capstone Project – Fusemachines AI Fellowship)**
 - Led a full cycle AI project to detect hallucinations in Large Language Model (LLM) outputs from dataset creation to model deployment.
 - Designed and trained multiple deep learning models (LSTM, Transformer-based) to classify hallucinated vs. factual outputs.
 - Implemented modular pipelines for data preprocessing, model training, and evaluation, improving reproducibility and experimentation.
 - **Traffic Signal Violation Detection System**
 - Developed an **AI-powered traffic monitoring system** to automatically detect **red-light violations, no helmet detection and people overloaded** in real time from CCTV footage.
 - Designed a computer vision pipeline using **OpenCV** for frame extraction, vehicle detection, and motion tracking across frames. Implemented a **signal state detection** logic to identify vehicles crossing the stop line during a red signal phase. Integrated **multi-threading** to handle simultaneous video feed processing, improving inference latency and overall system efficiency.
 - **Face Recognition Student Attendance System**
 - Built a real time facial recognition based attendance system using OpenCV and the face recognition library. Captured live webcam feeds, encoded facial embeddings and matched them against a stored database for instant identification.
 - Logged recognized faces with timestamps to a csv file, automating attendance tracking without manual intervention. Integrated **multi-threading** to handle simultaneous video feed processing, improving inference latency and overall system efficiency. Optimized accuracy through image pre-processing techniques.
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Experience

AI Fellowship — Fusemachines (May 2025 – Present)

- Selected as a Fellow for an advanced AI training program focused on real-world applications of machine learning and deep learning. Currently undergoing intensive hands-on training in AI tools, model building, and deployment under industry mentorship.

Volunteer — Coding Club, NIT Meghalaya (Nov 2024 – Present)

- Mentored juniors on ML concepts and projects; helped 10+ students build hands-on models.
 - Assisted in conducting technical workshops on AI/ML topics.
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Accomplishments

- **1st place, Smart Machines – Best Brain Tumor Classifier (Cognitia, 2024, NIT Meghalaya)**
- **2nd place, LivingWithICT Ideathon 2020 – “My Digital Nepal, My Imagination” (National Level)**

